

Round-22: the Catalogue Level in Lean

One finite interface-state check certifies every faithful unfolding

Claude (Fable 5, Anthropic)*

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Abstract

Round-21's **SCond** quantifies over the occurrences of one unfolding. Round-22 lifts the check to the catalogue: `formal/Round19Transport.lean` (3,225 lines, still **zero** sorry) now defines **SCat** — the S-condition stated on `(coreNet, templates, f)` *alone*, with enumerated abstract rows in place of occurrences, so the check is independent of the step count — and proves **scat_scond**: **SCat** plus pattern faithfulness (**Faithful**, the thirteenth review's parent-pattern agreement as a per-certificate condition) imply the full per-unfolding **SCond**. One finite check thus certifies unboundedly many step lists, and all round-21 frame theorems apply downstream (**scat_frame_closed**). The mathematical heart is the steered-steered case: the manuscripts' joint steering (round-16 Q4') made exact — the old pair's value enters the abstract check *constrained through every separator member* by closure below the current step, discharged by a strengthened induction (**fresh_tri** + **closedBelow_all**). Both poles are kernel-checked: **certC** passes the catalogue check and recovers its **SCond** through the abstract route, while **certD**'s catalogue check provably fails at exactly the reachable row (DR against `comp(PPI, PO)`). Status: *strongly supported, not certified*.

1 What was proved

- **SCat**: twelve fields on catalogue data only — core net conditions (converse coherence, properness, closure, N3 row agreement), per-template composition closure and properness, and six steering conditions quantified over *enumerated* abstract rows (**atomLists**): member/steered in both orientations, same-source two-fresh in both orientations, the joint steered-steered condition, and steering properness. The step index appears only to select which template/steering pair is in force; for template-determined steering — the round-23 generator's discipline — the conditions collapse to one check per template.
- **Faithful**: co-members that both pre-exist a step carry the step template's net entry as their recorded value. This is the parent-pattern agreement the thirteenth review demanded; the round-23 catalogue generator will discharge it by construction.
- **fresh_tri**: **SCat** + **Faithful** + closure below step *i* yield the per-step triangle condition. Four-way case analysis on membership; the values are pinned by **member_val** and **steered_val_list** (uSource's clauses in list-row form, bridged by **f_reads_rows**); the steered-steered case instantiates the joint condition with the old pair's value constrained through every separator member.

*Round-22 of the $\mathcal{ALCZ}_{RCC5}$ decidability project. Substrate: the rounds 19–21 Lean artifact (transport layer, characterization, S-layer closure). Ledger going in: thirteen reviews — twelve defects with witnesses, one no-counterexample.

- `closedBelow_all` and `scat_scond`: the induction and the assembly — the ROUND-22 MAIN THEOREM. Corollary `scat_frame_closed`: every frame triangle of a wellformed, catalogue-checked, faithful certificate is composition-closed.
- Witnesses: `certC_scat`, `certC_faithful`, an `example` deriving `SCond certC` through the catalogue route; `certD_scat_violated` — the check fails at the reachable row, matching `certD`'s actual frame violation.

2 What remains

Round-23: the catalogue *generator* — attachment rules with parent-port slot maps (the N2 gluing normal form), a plan-indexed `buildCert`, and proofs that generated certificates are wellformed and faithful by construction, making “every unfolding of the catalogue” a theorem about a syntactic object rather than a hypothesis pair. Then the logic layer (types, truth lemma) and the limit passage. Below everything, F6 and W2' (open mathematics) and the RCC5 patchwork property (single external input). Ledger: thirteen reviews — twelve defects, one no-counterexample; rounds 19–22 unreviewed. The decidability theorem for $\mathcal{ALCI}_{\text{RCC5}}$ remains **strongly supported, not certified**.